

Per-Stream Terminal Valence Closure:

A Boundary Hypothesis and Framework for Empirical Testing

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Abstract

Per-Stream Terminal Valence Closure (PTVC) presents a precise, falsifiable candidate boundary hypothesis at the intersection of affective science, consciousness research, measurement theory, stochastic processes, and endpoint modeling. On its primary finite signed-measure branch, it asks whether each separately routed maximal qualifying conscious stream carries exactly equal positive and negative Jordan masses across its complete strict-pre-endpoint domain, so that the terminal signed residual is zero. The equality is intrastream and nontransferable. This compact boundary condition is compatible with the full diversity of lived experience: lives may differ in gross valence variation, duration, intensity, order, agency, and structure.

$$Z_i^* := \nu_i(\Omega_i) = V_i^+ - V_i^- = L_i^{\text{phen}}(T_i^* -).$$

$$\text{PTVC}_i \Leftrightarrow V_i^+ = V_i^- < \infty \Leftrightarrow Z_i^* = 0.$$

The boundary condition becomes scientifically testable when joined to a prospectively fixed architecture for representation, sign, origin, scalarity, exposure, bearer, ownership and routing, domain, endpoint, observation and missingness, measured-to-latent recovery, reference class, and modality. Within that architecture, endpoint timing supplies a particularly informative structural discriminator for PTVC. On a prospectively declared positive-density endpoint branch under the at-risk law P_i^0 , suppose the endpoint probe is independent of the sigma-field generated by a jointly measurable pre-endpoint ledger path and has a conditional density strictly positive Lebesgue-almost-everywhere on a deterministic interval. If exact closure holds at the probe P_i^0 -almost surely, then, for P_i^0 -almost every eligible path, the ledger is zero Lebesgue-almost-everywhere on that interval. State-coupled timing, first hitting or absorption, endpoint atoms, singular carriers, routing change, continued experience, and endpoint-inclusive value define distinct, testable branches.

PTVC supports a research program structured for preregistration. Representation and origin validity, generator recovery, and observation coverage form its foundation; random-cut, signed-dose, horizon, reserve, and completion-potential studies probe candidate routes. On the declared action-first route, fixed-policy probability-one completion requires probability-one policy mass on actions whose complete frozen continuations close with probability one. This result characterizes probability-one policy mass on those actions; topological-support containment, psychological motivation, and endpoint support remain distinct protocol-fixed questions. Measurement, regulation, mood, consciousness, terminal-trajectory, missing-data, and stochastic-endpoint literatures supply instruments, comparators, data, and safe perturbations. PTVC-specific support requires frozen predictions to transport across held-out modules and outperform equal-budget adaptation, regulation, mood, bridge, selection, memory, observation, and hidden-state rivals. Finite-resolution compatibility, an atom at zero, probability-one closure, and set-universal closure remain distinct. Complete-stream inference requires complete-domain coverage plus validated endpoint and measured-to-latent bridges. The contribution joins a consequential conjecture to measurable signatures, discriminating tests, and adjudication of a possible terminal organizing principle of conscious experience.

Keywords: PTVC; Law of Fairness; valence; signed measures; terminal boundary condition; latent-variable identification; consciousness; endpoint processes; completion potential; affective science; partial identification; model recovery



Illustration by Diego Santos.

1. Scientific question, scope, and research opportunity

The scientific question is sharply defined: after independently fixing a qualifying bearer, exhaustive ownership and routing, a complete strict-pre-endpoint truth domain, prospectively fixed valence representation, experienced-exposure rule and clock, neutral origin, and truth-level endpoint, are the positive and negative Jordan masses finite and equal for every separately routed maximal qualifying stream in the declared class?

Prospective-specification convention. Protocol-fixed means written into the study protocol before outcome inspection. Prospectively specified means fixed before outcome inspection without asserting a public record. Publicly preregistered means deposited before outcome inspection in a timestamped external registry with a persistent identifier; cite that identifier when making the claim. Without it, use protocol-fixed or prospectively specified.

A boundary hypothesis with a precise domain

PTVC concerns one cumulative invariant while welfare, opportunity, agency, duration, remembered utility, social outcomes, and gross experience remain distinct dimensions. A stream containing large positive and negative magnitudes may satisfy the same terminal residual condition as a stream containing small magnitudes. This separation preserves the independent importance of rights, justice, suffering prevention, dignity, and practical welfare. Every application of the hypothesis must uphold treatment, protection from harm, and active concern for suffering.

What makes the claim scientific

Scientific force comes from prospectively fixed gates. The reporting architecture fixes the representation, bearer, domain, endpoint, observation, recovery, reference class, and modality before decisive analysis. PTVC must then outperform equally resourced origin-choice, stream-partition, endpoint-alignment, missingness, instrument-floor, and conditioned-model rivals. Each passed gate advances only its linked claim. An unmet gate blocks promotion beyond the last level supported by the evidence and identifies the next measurement, bridge, or recovery task.

Scientific foundation and open empirical question

PTVC is mathematically specified and empirically adjudicable under explicit measurement, bridge, and identification assumptions. Existing work on affective measurement, adaptation, opponent regulation, computational mood, consciousness, terminal trajectories, and missing data shows that many required components can be measured and modeled. This article states a precise target, derives discriminating implications, and supplies a preregistration-ready program for accumulating support, refining declared branches, and adjudicating them through prospective tests. Complete-stream empirical adjudication remains the program's culminating objective.

Interpretive context. Ideas of balance, restoration, consequence, and completion can generate comparative hypotheses across cultural narratives and lived experience. PTVC-specific scientific weight arises when prospectively specified terminal and cross-module predictions outperform cultural transmission, memory, comfort, cooperation, just-world belief, and other serious rivals.

Contribution and novelty

The article's distinctive contribution is the typed assembly of established signed-measure, stochastic-process, reachability, conditioning, identification, and model-recovery tools around the PTVC target, together with the branch-specific consequences derived from that assembly. The result is a novel, integrated research architecture: an exact intrastream terminal condition linked to random-endpoint, reachability, action-mixture, partial-observation, and transport tests. The displayed propositions remain conditional on named assumptions; optional routes receive route-level weight through their protocol-fixed bridges; and finite sampling reaches [SET-ALL] through an independent quantifier bridge.

2. Formal object and primary finite-measure claim

Fix a separately routed maximal qualifying stream i and its measurable complete strict-pre-endpoint truth domain (Ω_i, Σ_i) , and a prospectively declared signed phenomenal representation. On the primary branch, $v_i: \Sigma_i \rightarrow \mathbb{R}$ is a finite signed measure with Jordan decomposition $v_i = v_i^+ - v_i^-$ into mutually singular positive measures. V_i^+ and V_i^- are their finite Jordan masses; their difference is the terminal signed residual.

Primary finite signed-measure branch

$$v_i = v_i^+ - v_i^-, \quad v_i^+ \perp v_i^-. \\ V_i^\pm := v_i^\pm(\Omega_i), \quad V_i^{\text{gross}} := V_i^+ + V_i^- = |v_i|(\Omega_i).$$

$$Z_i^* := v_i(\Omega_i) = V_i^+ - V_i^- = L_i^{\text{phen}}(T_i^* -).$$

$$\text{PTVC}_i \Leftrightarrow V_i^+ = V_i^- < \infty \Leftrightarrow Z_i^* = 0.$$

The claim is exact, intrastream, and nontransferable: a residual in one stream cannot be canceled by experience in another. It is also compatible with nonzero gross variation. The mixed-sign case $V_i^+ = V_i^- > 0$ is therefore structurally distinct from a degenerate stream with no phenomenal exposure. Vector-valued, bipolar, nonadditive, or generalized terminal-neutrality branches remain possible, but they are not silently treated as the scalar additive claim. The universal quantifier is conditional on admissibility of the protocol-fixed finite representation; a broader all-stream claim requires a separate domain-coverage or branch-spanning result.

A latent ledger requires an exposure model

On an absolutely continuous subbranch, a latent ledger can be written as an exposure-weighted integral. The general display retains a left-limit term, but on the primary whole-stream branch S_i^θ is the truth-domain origin and $L_i^{\text{phen}}(S_i^\theta -) = 0$; later entry uses the delayed-entry decomposition on the half-open observed interval $[a, b)$. Here $g_{\theta,i}$ and q_i^{exp} are the two factors in the prospectively frozen exposure-density convention. On the primary convention, $g_{\theta,i}$ is a dimensionless time-chart, unity, or ownership factor and q_i^{exp} has exposure-per-time units; on the unity-clock alternative, $q_i^{\text{exp}} \equiv 1$ and $g_{\theta,i}$ carries exposure-per-time units. Their product is experienced-exposure density; neither factor may duplicate ownership or exposure already encoded by μ_i^{phen} , and v_i^{real} is realized neutral-relative phenomenal value per unit exposure.

Exposure-weighted latent ledger

$$L_i^{\text{phen}}(t) = L_i^{\text{phen}}(S_i^\theta -) + \int_{S_i^\theta}^t g_{\theta,i}(s) q_i^{\text{exp}}(s) v_i^{\text{real}}(s) ds.$$

Additivity is the declared representation commitment. Once a signed measure is defined, $v_i(\Omega_i) = v_i^+(\Omega_i) - v_i^-(\Omega_i)$ is an identity. PTVC adds the substantive boundary restriction that the independently defined residual equals zero at the complete endpoint. Constructed zero-integral functions establish mathematical coherence and help design controls; the empirical opportunity is to determine whether independently calibrated natural phenomenal histories satisfy the same boundary.

Primary target: a boundary property of a complete, independently defined stream, distinguished from normalization rules, population averages, and narrative judgments.

Bipolar and generalized branches

The net signed measure need not exhaust phenomenal structure. A bipolar branch allows positive and negative components to coexist, so a common overlap component can increase primitive gross magnitude without changing the net ledger. The net measure then identifies a sharp lower bound on gross variation but no finite upper bound unless overlap is independently bounded. A zero net residual therefore never implies a low-experience or low-suffering stream. Generalized Terminal Neutrality (GTN) is retained for nonadditive or functional-zero representations, but GTN is not relabeled as PTVC unless a valid bridge connects its zero to the additive or bipolar terminal object.

3. Quantifiers, ownership, routing, and falsification

Stream-level adjudication of the universal claim

The primary strict-universal formulation is [SET-ALL]: every separately routed maximal qualifying stream in the protocol-fixed truth-domain class satisfies exact closure. A single gate-valid stream whose nonempty truth-containing residual set—an assumption-guaranteed outer set containing the latent residual—excludes zero is a deterministic counterexample to that protocol-fixed universal branch. A frequentist confidence set supports only its declared coverage statement; a posterior probability is conditional on its protocol-fixed prior and model; and a point estimator remains subject to its uncertainty analysis. Population means, favorable subgroups, or transfers between streams cannot average away a verified counterexample. Probabilistic branches retain their declared probability and error guarantees.

Bearer and routing rules are causal, not clerical

The unit of analysis is a maximal qualifying conscious stream under a prospectively fixed bearer and ownership rule. Splitting one stream after observing its residual, merging streams, dropping weakly observed intervals, or duplicating an event can create apparent balance. Routing must be exhaustive, nonduplicative, time-consistent, and outcome-independent. If individuation is partially identified, report the set over all admissible routings. Unresolved positive and negative contributions enter separate nonnegative ownership-slack accounts and may not cancel each other. Population claims additionally require declared stream-level inclusion probabilities or survey weights; convenient observed streams do not define the target population. The protocol is fixed before quantifying over streams, histories, continuations, or policies.

Modal claims form a partly ordered family

Existence of a completing continuation, positive-probability completion, fixed-policy probability-one completion, almost-sure attainability, limit-sure attainability, topological-support containment, and closure of every admissible continuation are distinct. Almost-sure attainability requires an identified attaining policy; limit-sure attainability permits only a sequence whose values approach one (Asadi et al., 2025; Chatterjee et al., 2016). Every experiment must freeze its modality, intervention or policy, continuation class, and evidence object. None of these claims may be promoted to [SET-ALL] without a separate quantifier bridge.

Target and route are separate axes

Target support concerns the terminal residual under a validated bridge. Route support concerns an optional mechanism proposed to generate or constrain closure. Their separation strengthens inference: route failure leaves the target open, while route fit advances the target verdict only through a proved entailment.

Evidence remains branch-matched: exact-zero, finite-band, probability, and mechanism claims each use their corresponding support, rejection, and abstention rules.

Decision status remains typed

Public reporting preserves the Formal Model’s four-axis public result vector—TargetStatus, RouteStatus, ComparatorStatus, and InferenceStatus—together with a supporting record of prerequisites, target specification, recovery, and attainment. The axis vocabularies and controlled summary labels are those fixed in the synchronized Formal Model; this article introduces no parallel verdict lexicon. No composite score may override an adverse target result, rescue a failed prerequisite, conflate target geometries or modalities, or turn route superiority into proof of the terminal law.

4. Representation, neutral origin, and delayed entry

Neutral origin must be calibrated independently

A scalar ledger requires a sign convention, cross-sign comparability, an exposure unit, and a neutral origin. These are measurement hypotheses. Under the admissible affine recoding $v' = a_i + b_i v$ with $b_i > 0$, a stream-constant positive scale preserves a zero residual, whereas on positive-exposure records an additive shift preserves zero only when $a_i = 0$. The origin must therefore be calibrated on nonterminal data or bounded as an uncertainty set before the terminal residual is inspected. The admissible origin class—basis, complexity, envelope, and calibration sample—must be frozen. If it can select a record-specific function that forces zero, the origin is unidentified and the confirmatory result is No adequate estimate (NAE).

Origin sensitivity under affine transformation

$$\tilde{Z}_i^* = a_i E_i^{\text{tot}} + b_i Z_i^*, \quad b_i > 0, \quad E_i^{\text{tot}} = \mu_i^{\text{phen}}(\Omega_i) < \infty.$$

If total exposure is positive, choosing $a_i = -b_i Z_i^* / E_i^{\text{tot}}$ forces the transformed residual to zero. That operation is precisely why completed-record mean-centering is prohibited as confirmatory evidence. A valid cross-sign program perturbs independently validated positive and negative channels, estimates their exchange map, tests state dependence and hysteresis, and freezes the admissible origin set before endpoint fitting.

Study entry and stream origin are distinct

Because most studies start late and stop before the true endpoint, whole-stream claims require one joint truth-containing prefix–observed–suffix object on the same latent scale. It must be narrow enough to decide the protocol-fixed exact or finite-band target; separate component bounds become decisive only through that joint object.

Delayed-entry decomposition

$$\begin{aligned} Z_{i,\text{pre}}(a) &:= L_i^{\text{phen}}(a -), & Z_{i,\text{obs}}([a, b]) &:= L_i^{\text{phen}}(b -) - L_i^{\text{phen}}(a -), \\ Z_{i,\text{suf}}(b) &:= L_i^{\text{phen}}(T_i^* -) - L_i^{\text{phen}}(b -), & Z_i^* &= Z_{i,\text{pre}}(a) + Z_{i,\text{obs}}([a, b]) + Z_{i,\text{suf}}(b). \end{aligned}$$

The three terms must be constructed jointly so their dependence and simultaneous guarantee are preserved. If either unobserved end is unbounded or no valid joint truth-containing set can be constructed, the complete-stream result is No adequate estimate (NAE). If a valid joint outer set intersects both the protocol-fixed acceptance set and its complement, the result is indeterminate. A measured branch applies the same accounting to its declared scale without silently promoting it to the latent phenomenal ledger.

Identification gate: a stable sign, origin, unit, and cross-sign scalarization promotes the scalar branch; if one component remains unresolved, the vector-valued alternatives stay available for corresponding tests.

Factorization, scalarity, and gauge freedom

On a density branch, phenomenal contribution can be factorized into exposure and neutral-relative value per unit exposure. Positive predictable rescaling of one factor with inverse rescaling of the other leaves the signed measure unchanged, so exposure and intensity are not separately identified by their product. Clock time, subjective duration, attention, consciousness density, and unity weights must be compared under independent manipulations rather than selected for terminal fit. Raw ratings, ranks, embeddings, factor scores, and logits are not automatically cardinal. Additive-conjoint independence or separability, cancellation, solvability, and held-out consistency test whether one cross-sign scalar is defensible (Luce & Tukey, 1964); bipolar co-occurrence tests whether positive and negative components must instead coexist (Cacioppo & Berntson, 1994; Larsen et al., 2001). An adequate reproducible violation fails the scalar prerequisite; nonidentification or invalidity yields NAE. Prespecified vector, ordered, set-valued, and GTN branches remain open. That representation result is not adverse TargetStatus evidence for every branch.

5. Observation, exactness, and the inference ladder

Let P_i^Z denote the latent terminal-residual law on the declared truth space. Observed reports, physiology, behavior, neural signals, and clinical records arise through a Markov kernel $K_{i,\eta}^{\text{obs}}$ from that latent space to a specified observation space; η indexes instrument, resolution, care context, communication, sedation, cadence, retention, and other declared conditions. The observed law $P_{i,\eta}^{\text{obs}}$ is the kernel push-forward of P_i^Z . Observed variance contraction therefore need not be latent contraction.

Observation-kernel push-forward

$$P_{i,\eta}^{\text{obs}}(A) = \int K_{i,\eta}^{\text{obs}}(A | z) P_i^Z(dz).$$

Target geometry and exactness levels

- Latent exact target: for a nonempty truth-containing residual set R_i , exact-compatible means $R_i = \{0\}$; exact-adverse means zero is not an element of R_i ; every other relation is exact-indeterminate. A confidence set containing zero is not an exact truth-containing singleton.
- Finite-band target: for a protocol-fixed scientific acceptance set, band-equivalent means the truth-containing residual set is wholly inside it; band-adverse means the two sets are disjoint; partial overlap is band-indeterminate. The familiar $[-\varepsilon, \varepsilon]$ band is one scalar special case. A law-level band-probability claim must separately fix its stream or population law, minimum mass threshold, and error rule in the protocol before outcome inspection.
- Exact atom: the latent terminal law assigns positive mass to zero. Shrinking observed bands do not establish an atom without a validated inverse bridge, deconvolution, or partial-identification argument.
- Law-level quantifiers: probability-one closure under a declared random-stream law and set-universal closure over every qualifying stream are distinct. Neither follows from a large estimated atom mass or a finite sample of compatible streams.

Inference must preserve the object type

Scientific acceptance geometry and epistemic uncertainty are different objects. The acceptance set is fixed prospectively; uncertainty widens the identified region, truth-containing outer set, confidence set, credible set, or sensitivity region rather than that acceptance set (Manski, 2003). A frequentist confidence set is a random set whose protocol-fixed procedure controls repeated-sampling coverage; it is not a deterministic truth-containing set. A Bayesian decision remains prior- and model-conditional, while a deterministic falsifier requires a truth-containing set. Origin, endpoint, routing, missingness, kernel, and numerical uncertainty must be propagated. No geometric relation silently converts finite-band evidence into exact or universal closure.

Equivalence requires affirmative evidence

Equivalence is supported through prospectively specified margins, family-wise error control, boundary-aware asymptotics, mixture alternatives, and model-recovery performance rather than through nonsignificance alone (Schuirmann, 1987; Lakens, 2017; Self & Liang, 1987). A nonidentifying observation map keeps inference at the measured scale until a valid latent bridge is supplied.

Exact atoms require stable inverse recovery

An injective observation map on a protocol-fixed law class does not by itself guarantee stable finite-sample recovery of mass at exactly zero. Diffuse laws concentrated arbitrarily near zero can resemble an atom after noise, rounding, censoring, smoothing, or kernel uncertainty. Recovery studies must therefore include true atoms, near-zero diffuse alternatives, locally discrete laws without a zero atom, mixtures, singular components, quantized channels, observation floors, and perturbed kernels. If the pipeline cannot distinguish those generators at the protocol-fixed resolution, the proper conclusion is a finite band, partial identification, or No adequate estimate. Under a locally smoothing observation kernel, atomless latent laws can concentrate arbitrarily near zero and become arbitrarily close in observed total variation to a true zero atom at every fixed

sample size; exactness therefore requires a protocol-fixed separation or structural restriction, not merely more observations under the same unresolved kernel.

6. Random endpoints: a sharp structural discriminator

Exact closure at an independently timed endpoint with a density positive Lebesgue-almost-everywhere imposes a strong and informative terminal signature. Under the assumptions below, for almost every eligible path under the declared conditional reference law, the pre-endpoint ledger equals zero for Lebesgue-almost every time on the interval.

Independent positive-density endpoint proposition

Let E_i be a prospectively declared positive-probability event on which the stream, frozen routing and truth domain, finite ledger, and pre-killing extension exist on a deterministic interval I ; write $P_i^0 = P(\cdot | E_i)$. Under this same law, let $\tilde{L}_i(t-)$ be jointly measurable, $\mathcal{F}_i^1 = \sigma\{\tilde{L}_i(s-): s \in I\}$, and T_i^{exo} independent of $\mathcal{F}_i^1$. Assume T_i^{exo} is I -valued, has a Lebesgue density positive almost everywhere on I , and $P_i^0\{\tilde{L}_i(T_i^{\text{exo}}-) = 0\} = 1$.

Independent positive-density endpoint obstruction

$$0 = P_i^0(\tilde{L}_i(T_i^{\text{exo}}-) \neq 0) = \int_I P_i^0(\tilde{L}_i(t-) \neq 0) f_{T_i^{\text{exo}}}(t) dt.$$

The integrand is nonnegative. Tonelli's theorem and the positive density give the product-measure statement: $P_i^0\{\tilde{L}_i(t-) \neq 0\} = 0$ for Lebesgue-almost every $t \in I$; Fubini's section conclusion equivalently gives $\tilde{L}_i(t-) = 0$ for Lebesgue-almost every t on a P_i^0 -probability-one set of paths. If paths are left-continuous P_i^0 -almost surely, equality holds at every interior time; the left endpoint also requires right continuity or a separate value condition. Thus, on the independent positive-density branch, exact closure requires the zero-almost-everywhere structure identified by the proposition.

What the discriminator reveals

The proposition identifies the structure a nondegenerate implementation must supply. Candidate implementations include state-coupled endpoint timing, first-hitting or absorbing boundaries, endpoint atoms or singular support, phenomenally active endpoints, routing change, and failure of the pre-killing extension. Each defines a testable branch with distinctive predictions.

Empirical use

Unexpected and weakly anticipated endpoints are high-pressure tests. The analysis must measure residual dependence between the path and endpoint after prospectively declared covariates, report endpoint surprise, and compare state-coupled hazards, selection, observation shutdown, and random-endpoint bridge rivals. An unplanned endpoint is not thereby exogenous; the protocol-fixed dependence, surprise, and rival checks provide a graded sensitivity analysis, while the independent-endpoint proposition applies only when its independence premise is separately justified.

What the proposition establishes and which branches it opens

The branch-specific discriminator prospectively localizes routes. Every nontrivial exact-closure model must show structure beyond independent positive-density timing; the candidates above make that testable. Premises remain fixed before residual inspection. Approximate independence gives graded pressure when endpoints sample nonzero-ledger times. Ledger-dependent timing identifies a route for direct evaluation; the terminal test adjudicates closure.

Approximate dependence yields a graded pressure test

For $\varepsilon \geq 0$, define $X_i := (\tilde{L}_i(t-))_{t \in I}$, let C_i be prospectively frozen pre-endpoint covariates, and fix a Borel set $J \subseteq I$ with $0 < \lambda(J) < \infty$, where λ is Lebesgue measure. Under P_i^0 , define $g_{i,\varepsilon}(t, c) := P_i^0(|\tilde{L}_i(t-)| > \varepsilon | C_i = c)$, $p_{i,\varepsilon} := P_i^0(|\tilde{L}_i(T_i^{\text{probe}}-) | > \varepsilon)$, and $\rho_i := E_{P_i^0}[d_{\text{TV}}(P_{X_i, T_i^{\text{probe}} | C_i}, P_{X_i | C_i} \otimes P_{T_i^{\text{probe}} | C_i})]$, where the conditional kernels are evaluated at C_i and $d_{\text{TV}}(P, Q) := \sup_A |P(A) - Q(A)|$. Assume standard-Borel path and covariate spaces, measurable evaluation, the required regular conditional laws, and $f_{T_i^{\text{probe}} | C_i}(t | C_i) \geq m_J > 0$ for λ -almost every $t \in J$, P_i^0 -almost surely.

Approximate endpoint-dependence occupation bound

$$E_{P_i^0}[\int_J g_{i,\varepsilon}(t, C_i) dt] \leq \frac{p_{i,\varepsilon} + \rho_i}{m_J}, \quad m_J > 0.$$

The total-variation event bound changes the probe off-band event by at most p_i ; the density floor then gives the display. Consequently, substantial off-band occupation together with rare probe off-band events requires correspondingly large endpoint-ledger dependence after the declared covariate adjustment. Its scientific role is to quantify sensitivity to dependence; independence and SET-ALL remain separate protocol-fixed analyses.

7. Reachability, endpoint support, and solvency

A residual must be cancelable by an admissible future

For a realized stream-owned prefix h_t and prospective endpoint u , define the admissible continuation class $C_i(t, u | h_t)$ without terminal-outcome leakage. Let Cap_i^+ be the supremum attainable future positive contribution and Cap_i^- the magnitude of the infimum attainable future negative contribution. Any continuation that closes the ledger must contribute the negative of its current value, yielding the necessary directional solvency screen:

Directional reachability necessity

$$-\text{Cap}_i^+(t, u | h_t) \leq L_i^{\text{phen}}(t) \leq \text{Cap}_i^-(t, u | h_t).$$

A residual outside this interval makes exact completion at u impossible. An empty continuation class also makes completion impossible, and no capacity supremum is evaluated. An undefined, invalid, or unrecoverable class or decision object yields unavailable or NAE. Passing establishes capacity feasibility. Joint attainability then follows from compactness or closure, measurable selection, and an existence result, while the protocol-fixed diagnostics address infinite or unattained suprema (Aubin, 1991).

Endpoint support localizes the required structure

If endpoints can occur over broad support, the residual must remain solvent there. Suppose the endpoint counting process has a predictable, locally integrable intensity on the at-risk set that generates an absolutely continuous compensator of finite mass on the analysis interval; assume no simultaneous ledger–endpoint jump. Intensity weighting cannot create a new exact atom at zero from an occupation law that is atomless there. Exact atomic closure instead requires pre-existing atomic occupation, absorption or stickiness, a first-hitting construction, a predictable endpoint atom, a singular compensator, or a justified limiting mechanism. Atomic, singular-compensator, first-hitting, and limiting branches remain separate.

Endpoint adjudication is versioned evidence

Candidate crossing, Estimated endpoint, Confirmed under protocol, Revised endpoint, or No adequate estimate are distinct statuses. Dwell time, return logic, competing events, reliability, and revision rules are frozen before ledger inspection. Later recovery or reclassification preserves the original analysis version and propagates endpoint uncertainty; the last available observation is never forced to be the truth-level endpoint.

Reachability measures future capacity; actual selection, mechanism, and completed closure are evaluated through their corresponding protocol-fixed tests.

Endpoint branches are empirical alternatives

Independent positive-density timing predicts the obstruction above. First hitting or absorption predicts hazard concentration on a zero surface. A predictable endpoint atom permits exact mass at a scheduled or discrete boundary. A singular carrier requires carrier-specific occupation tests. An endpoint-inclusive branch requires independently supported phenomenal mass and ownership at the boundary. A continued-stream branch separates biological death from the truth endpoint. Split, fusion, or routing change requires prospective junction rules and nonduplication. Each branch has different data requirements and can be retired; none is a post-outcome exemption. A first-hitting or absorption model supplies route evidence only when the stream endpoint is defined independently of the ledger and its predicted coincidence with the hit is tested; defining the endpoint as a zero hit makes closure true by construction. On an optional biological-death-bounded branch, the phenomenal endpoint is no later than independently adjudicated biological death, but the biological event remains a distinct endpoint model and is not automatically the last qualifying experience.

Unexpected or weakly anticipated natural endpoints are especially informative because a local controller may lose sufficient actionable horizon while a global or state-coupled route may transport. Ethical use is observational: no study may alter survival or care to create endpoint variation.

8. Optional mechanism routes and path-class constraints

PTVC is deliberately mechanism-neutral, allowing reduced reserve-coupled routes, conditioned bridges, absorbing processes, constrained controllers, and other constructions to compete through their distinctive assumptions and predictions. The core target remains independently evaluable across those routes.

Finite variation restricts exact diffusion bridges

A finite signed-measure ledger has finite variation; a continuous version therefore has zero continuous quadratic variation. Suppose an exact path bridge identifies it with a reserve-coupled semimartingale whose continuous local-martingale

coefficient is the product of a positive ledger–reserve conversion constant, the remaining reserve, and the diffusion coefficient, while the remaining terms have finite variation. Writing continuous quadratic variation in the displays below, exact path equality requires both increments to vanish:

Finite-variation obstruction to an exact diffusion bridge

$$\begin{aligned} [L_i^{\text{phen}}]_t^c - [L_i^{\text{phen}}]_{S_t^\theta}^c &= 0. \\ [L_i^{\text{rc}}]_t^c - [L_i^{\text{rc}}]_{S_t^\theta}^c &= c_{\text{LU}}^2 \int_{S_t^\theta}^t U_i(s)^2 \sigma_i(s)^2 ds = 0. \end{aligned}$$

Here the route-specific remaining reserve and diffusion coefficient are assumed locally square-integrable with respect to the product of Lebesgue time and the governing reference law on the declared interval. Exact path equality then forces their product to vanish almost everywhere for that product measure; wherever reserve remains positive, the diffusion coefficient must vanish in the same sense. A nondegenerate diffusion may remain an approximate, terminal-only, or consequence bridge, but not an exact continuous finite-variation path identity without resolving this mismatch (Protter, 2005). Jumps require separate quadratic-variation accounting.

Reserve and endpoint require a tested bridge

Mechanism-level reserve inference begins with a proxy distinct from ordinary time-to-endpoint prediction. A credible proxy is constructed without terminal-neutrality labels, adds predictive value beyond prognosis, frailty, disease severity, treatment, care setting, device dropout, and observation availability, and transports across sites and perturbations. A separately tested coincidence bridge can then connect route-state exhaustion with the truth-level stream endpoint.

Mechanism-localization discipline

An intervention may change assignment, information, policy, action, physical transition, outcome, retention, endpoint, routing, or observation. The earliest altered conditional law is the supported causal locus. When treatment, exposure, censoring, and endpoint intensity evolve in continuous time, causal weighting additionally requires positivity and a declared local-independence structure (Røysland, 2011). Downstream consequences of one alteration form one evidence family and may not be counted as independent confirmations.

Action-mediated route

Under the protocol-fixed action-first factorization, suppress the fixed protocol index and use the article-level alias below. Here the completion event denotes the induced joint action–suffix event under that fixed factorization; protocol and product-space indices are suppressed only for compression. Assume a measurable action-specific completion probability and probability-one policy mass on the structural action domain. Fixed-policy probability-one completion then holds exactly when the completion probability equals one for policy-almost every selected action. This result characterizes when the policy assigns probability one to the measurable set of actions whose completion probability equals one; topological-support containment, psychological motivation, and endpoint support remain distinct protocol-fixed questions.

$$\begin{aligned} p^{\text{comp}}(a \mid h, g) &:= p_{\eta, h, g}^{\text{comp}}(a), \\ \pi(\mathcal{A}^{\text{do}}(h) \mid h, g) &= 1, \\ P^{\pi, g}(F_{\text{comp}} \mid h) &= \int_{\mathcal{A}^{\text{do}}(h)} p^{\text{comp}}(a \mid h, g) \pi(da \mid h, g), \\ P^{\pi, g}(F_{\text{comp}} \mid h) = 1 &\Leftrightarrow p^{\text{comp}}(a \mid h, g) = 1 \quad \pi(\cdot \mid h, g)\text{-almost everywhere.} \end{aligned}$$

Exact-differential route

A separate route treats the ledger as a path integral of a one-form on a sufficiently rich state space. If the one-form is the differential of a potential Φ on the relevant domain, an open path from S to T has an endpoint-dependent integral, whereas every admissible closed loop has zero integral. Hysteresis, nonzero circulation, or path-dependent residue under frozen coordinates rejects that exact route unless the declared state is enlarged and independently revalidated.

Exact-differential route and loop diagnostic

$$\begin{aligned} \omega = d\Phi &\Rightarrow \int_{\gamma_{S \rightarrow T}} \omega = \Phi(x_T) - \Phi(x_S). \\ \oint_{\Gamma} \omega &= 0. \end{aligned}$$

Empirical weight comes from held-out route predictions beyond generic mathematical existence, including the generic ability to condition an ordinary process on completion; the attainability distinctions in Section 3 remain in force.

9. Completion potential and the transport fingerprint

A discriminating near-term route is a prospectively frozen completion potential transported across independent modules.

Completion potential

Let $\mathcal{H}_{i,t}$ be the frozen pre-endpoint filtration completed under P_i^0 , with $\mathcal{E}_{i,t}$ and the current-risk indicator $\mathbf{1}_{\{T_i^* > t\}}$ measurable with respect to $\mathcal{H}_{i,t}$. For positive-probability $A_{i,\varepsilon}$, require on $\{T_i^* > t\}$ that $P_i^0(A_{i,\varepsilon} \mid \mathcal{H}_{i,t}, T_i^* > t) = P_i^0(A_{i,\varepsilon} \mid \mathcal{E}_{i,t}, T_i^* > t) = h_{i,\varepsilon}(t, \mathcal{E}_{i,t})$ almost surely; otherwise augment the state or retain history. Nested bands require potentials monotone in ε . The shrinking-band limit is an exact-atom statement only under the stated regular-conditional construction.

Positive-probability completion potential

$$\begin{aligned} A_{i,\varepsilon} &:= \{ |Z_i^*| \leq \varepsilon \}. \\ h_{i,\varepsilon}(t, \xi) &:= P_i^0(A_{i,\varepsilon} \mid \mathcal{E}_{i,t} = \xi, T_i^* > t). \\ \lim_{\varepsilon \downarrow 0} h_{i,\varepsilon}(t, \xi) &= P_i^0(Z_i^* = 0 \mid \mathcal{E}_{i,t} = \xi, T_i^* > t). \end{aligned}$$

Let E_i^0 denote expectation under P_i^0 . Require $P_i^0(A_{i,\varepsilon} \mid \mathcal{H}_{i,0}) > 0$ almost surely on the declared risk set. The resulting process is a generic nonnegative normalized conditional-probability martingale and supplies calibration checks. PTVC-specific route evidence arises when protocol-fixed held-out relations transport beyond this generic behavior. Its declared domain requires the stated positive-probability condition; probability-zero conditioning uses a declared regular-conditional bridge, density limit, or other justified construction (Doob, 1957; Chetrite & Touchette, 2015).

Normalized completion-event martingale

$$\begin{aligned} M_{i,\varepsilon}(t) &:= \frac{P_i^0(A_{i,\varepsilon} \mid \mathcal{H}_{i,t})}{P_i^0(A_{i,\varepsilon} \mid \mathcal{H}_{i,0})}. \\ E_i^0(M_{i,\varepsilon}(t) \mid \mathcal{H}_{i,s}) &= M_{i,\varepsilon}(s), \quad 0 \leq s \leq t. \end{aligned}$$

The overidentification fingerprint

Estimate one potential on training modules without test-endpoint leakage; freeze its state, normalization, tolerance family, support, and observation pipeline; then predict held-out transition dynamics, action odds, endpoint support, terminal-band behavior, and resolution transport. Observational completion potential under P_i^0 and interventional completion potential under a declared policy are different estimands and may not be exchanged without identification and transport assumptions.

A Doob transform provides a generic conditioned-law benchmark; held-out transport beyond adequate conditioned rivals supplies PTVC-specific route support.

Conjunctive differential signature

Specificity comes from a conjunctive pattern of independently tested signatures. The primary proposed conjunctive route package requires reserve-dispersion behavior; two-time covariance and perturbation response beyond bridge rivals; endpoint-surprise calibration; signed-dose compensation with assigned, delivered, experienced, and phenomenal dose kept distinct; residual incremental value; residual compression sufficiency; transport across transition, action, endpoint, terminal-support, resolution, and observation modules; and equal-budget rival failure on common support. Incremental value and compression remain separate claims.

Random-cut complementarity

At a prospectively randomized cut, finite additivity links the accrued prefix and future remainder to the complete residual. The fixed disjoint half-open truth-domain slices assign any cut-time atom once. The identity is bookkeeping; PTVC adds an independently tested zero restriction. The observed remainder must come from post-cut data or a validated terminal-remainder bridge, never subtraction of the measured prefix from the tested terminal target. Use a two-stage within-stream design with randomized cuts, independent prefix and remainder calibration, stream-specific intercepts, cluster-aware uncertainty, and the protocol-fixed errors-in-variables model below. In the display, u_{ik} and v_{ik} are the protocol-fixed prefix and remainder observation or bridge errors; their joint law across cuts and channels is prospectively specified. Estimate β freely: $\beta = 1$ is the additivity gate, equivalently a protocol-fixed prefix coefficient of -1 . On the exact direct-partition branch, $\beta = 1$, and $\alpha_i = Z_i^*$ follows only when the relevant combined error location is fixed at zero by design or independent calibration. These are structural restrictions, not observed-data identification results. Interpret or test α_i only after a valid errors-in-variables

route identifies or bounds the joint error location, scale or covariance, and admissible bias through repeated measurement, validation data, instruments, structural restrictions, or bounded partial identification. Any additional latent residual belongs to a separately protocol-fixed bridge or model-misspecification branch, not the exact ledger identity. A bounded-episode zero intercept is local unless the cut partitions the complete qualifying stream or a protocol-fixed prefix–observed–suffix bridge covers the omitted domain; a pooled zero is a preliminary result that advances through stream-specific adjudication. Repeated cuts are dependent within stream and do not replace stream-specific adjudication. The slope gate establishes additivity; the independently tested terminal restriction supplies the corresponding PTVC test.

Random-cut additivity and stream-specific residual adjudication

$$\begin{aligned} Z_i^* &= L_{ik}^{\text{pre}} + L_{ik}^{\text{post}}. \\ \hat{L}_{ik}^{\text{pre}} &= L_{ik}^{\text{pre}} + u_{ik}, \\ \hat{L}_{ik}^{\text{post}} &= \alpha_i - \beta L_{ik}^{\text{pre}} + v_{ik}. \end{aligned}$$

10. Rival tournament under equal information budgets

Every candidate and rival receives the same raw variables, observation kernel, missingness inputs, data split, tuning, representation complexity, computation, and predictive information budget (Blackwell, 1953). Prospectively chosen proper scores compare predictive laws; the multi-gate decision rubric stays separate. Relative rank is followed by absolute-adequacy, recovery, common-support, and open-set checks. Table 1 gives the rival bank and pressure tests.

Table 1. Equal-information-budget rival models and their primary differential pressure tests.

Rival	Can mimic	Primary differential pressure test
Regulation / robust perfect adaptation	Adaptation, opponent, homeostatic, allostatic, or robust set-point return	Held-out signed integral and terminal residual beyond local set-point recovery
Computational mood	Recent expectations and prediction errors create history dependence	Frozen completion potential adds cross-module predictive structure
Flexible baseline	Nonzero subject- or state-specific terminal attractor	Compare zero-restricted and nonzero-attractor baselines under held-out proper scoring, calibration, and absolute-adequacy checks
Conditioned bridge	Endpoint conditioning pins paths by construction	Score event probability, normalizer, and unexpected endpoints
Endpoint selection	Hazard depends on latent state, frailty, care, or prognosis	Joint hazard–ledger fit and endpoint-surprise transport
Observation floor	Resolution loss, clipping, sedation, or channel shutdown	Independent channels and calibrated latent deconvolution
Memory kernel	Peak–end and duration-neglect reconstruct retrospective balance	Prospective experience ledger retains signal after memory adjustment
Expected neutrality	Population or conditional mean is zero with residual variance	Variance/atom/pathwise tests distinguish mean from closure
Open-set generator	Unmodeled dynamics fit none of the candidates	Absolute adequacy and no-forced-choice recovery
Ordinary policy control	Reward, effort, habit, threat, horizon, and viability shape action	Residual term adds held-out calibration beyond equal-information ordinary policy models

Let Q be one fixed prospectively declared rival law and let $\mathcal{G}_{i,0}^{\text{ex}}$ be its baseline information sigma-field. Expected neutrality under Q is a close but weaker balance claim. With a finite second moment, conditional mean zero together with positive conditional variance on a set of positive Q -probability does not imply an atom at zero or streamwise closure.

Expected neutrality is weaker than pathwise closure

$$\begin{aligned} E_Q[(Z_i^*)^2] &< \infty, & E_Q(Z_i^* | \mathcal{G}_{i,0}^{\text{ex}}) &= 0 \quad Q\text{-a.s.}, \\ Q(\text{Var}_Q(Z_i^* | \mathcal{G}_{i,0}^{\text{ex}}) > 0) &> 0. \end{aligned}$$

Model ambiguity is an admissible result. If two mechanisms induce the same distribution of protocol-observable variables under every allowed intervention, the study cannot identify their locus. The study remains nonidentifying until a new intervention or measurement separates them.

Generator recovery and verdict recovery are separate. A pipeline may rank the true generator yet issue the wrong scientific verdict or abstain correctly while generator identity remains ambiguous. Sealed simulations therefore score generator classification, favorable and adverse verdicts, model ambiguity, open-set detection, and no-adequate-estimate calibration separately; none can compensate for another.

11. Existing evidence and the scientific opportunity

Established neighboring results

Positive and negative affect can be partly separable and co-occur, supporting signed, bipolar, or vector representations (Cacioppo & Berntson, 1994; Larsen et al., 2001).

Adaptation after life events is well documented and heterogeneous (Diener et al., 2006; Luhmann et al., 2012). Opponent-process, allostatic, incentive-salience, interoceptive-prediction, homeostatic reinforcement-learning, computational-mood, and robust integral-feedback systems generate rebound, history dependence, dissociations among liking, wanting, and learning, and measurable forms of compensation (Solomon & Corbit, 1974; Koob & Le Moal, 2001; Berridge et al., 2009; Barrett & Simmons, 2015; Keramati & Gutkin, 2014; Eldar et al., 2016; Keren et al., 2021; Barkai & Leibler, 1997; Briat et al., 2016; Aoki et al., 2019). Together, these established dynamics show that valence is regulated across time through multiple interacting processes and provide biologically and computationally grounded route candidates and equal-budget rivals for PTVC.

Experienced and remembered utility can diverge through duration neglect and peak–end weighting (Fredrickson & Kahneman, 1993; Kahneman et al., 1997). This divergence makes prospective measurement and explicit memory rivals essential. PTVC extends that foundation into a new integrative question: whether complete-stream dynamics exhibit an exact terminal boundary condition.

Plausibility signals and evidential roles

- Hedonic adaptation, biological set-point control, and terminal change provide mechanism precedents, comparator models, and endpoint-study opportunities whose PTVC-specific weight is determined by protocol-fixed, branch-matched bridges.
- Constructed zero-integral functions, completion potentials, bridges, and Doob transforms establish mathematical coherence and generate testable predictions; held-out transport determines their empirical weight.
- Retrospective life evaluation provides informative evidence through a memory-and-demand observation kernel and can be compared directly with prospective experience ledgers.
- Endpoint proximity and changing channel variance become informative when modeled jointly with frailty, treatment, sedation, device failure, censoring, and observation floors.

How independent convergence builds confidence

The neighboring literatures contribute premises, instruments, comparator mechanisms, datasets, perturbations, and bias structures. A dependency graph makes shared samples, instruments, preprocessing, cultural ancestry, and observation channels visible so independent convergence receives appropriate weight. Confidence grows most when one frozen relation transports through distinct paths with different expected biases: prospective experience and retrospective memory; human and nonhuman tasks; survey, physiology, behavior, and neural measurement; ordinary and terminal horizons; observational cohorts and randomized perturbations. This triangulation strategy gives more weight to methodologically independent evidence than to raw citation count (Lawlor et al., 2016; Munafò & Davey Smith, 2018).

Endpoint, horizon, and consciousness literatures

Perceived future time reorganizes motivation and emotional priorities, making disclosed horizon a strong planning rival to actual endpoint coupling (Carstensen et al., 1999; Carstensen, 2006). Time-to-death studies report heterogeneous changes in affect, symptoms, depressive burden, and psychological or social well-being near death (Vogel et al., 2013; Seow et al., 2011; Petkus et al., 2025; Wang et al., 2025). These findings directly support endpoint-aligned study design and provide important baselines for the next PTVC-specific step: measuring a complete phenomenal ledger and testing terminal zero.

Behavioral nonresponse can conceal cognition; last report, last response, biological death, and last qualifying experience need not coincide (Casali et al., 2013; Bodien et al., 2024; Parnia et al., 2023). Comparative consciousness research cautions against limiting the eligible class to verbal humans (Birch et al., 2020). These literatures justify multimethod endpoint intervals with explicit uncertainty and provide concrete tools for testing stream individuation and eligibility across species.

Recent multicenter multimodal work in disorders of consciousness strengthens the case for combining behavioral assessment with neuroimaging and electrophysiological markers. Extending these tools with dedicated phenomenal-valence measures creates a credible path toward PTVC-relevant endpoint inference and makes the endpoint of qualifying conscious experience an explicit multimodal target (Manasova et al., 2026).

12. Measurement, missingness, and endpoint validity

Representation before accumulation

A confirmatory instrument must establish sign, origin, scale behavior, temporal resolution, exposure weighting, and measurement invariance across the populations and states being compared (American Educational Research Association et al., 2014; Krantz et al., 1971; Meredith, 1993). Method-diverse channels should be treated as separate fallible measurements of a latent object, not averaged until their error structure is modeled. Multitrait–multimethod validation and negative-control channels test whether common arousal, salience, motor capacity, or demand explains apparent convergence (Campbell & Fiske, 1959).

Observation coverage completes the estimand

Scheduled, randomized, participant-triggered, clinician-triggered, safety-triggered, endpoint-triggered, and device-failure observations retain separate marks. Crisis-enriched or terminally enriched samples alone do not identify a time- or exposure-weighted complete-stream integral.

Recovery requires an outcome-independent coverage component, a justified sampling-process correction with positivity, scientifically valid bounds for unsampled intervals, or an explicit partial-identification verdict.

Missing not at random is a central rival

Communication failure, fatigue, delirium, sedation, symptom severity, withdrawal, care transitions, and device loss can depend on latent valence and endpoint proximity. A 12-month burst-sampling study demonstrates that longitudinal completion and attrition patterns can be modeled explicitly and supplies a useful design template for the present missingness analysis (Prochnow et al., 2025). Selection, pattern-mixture, joint longitudinal-event, inverse-probability, and bounding approaches should be compared across prospectively specified sensitivity ranges rather than relying on complete-case or last-observation-carried-forward defaults (Rubin, 1976; Scharfstein et al., 1999; Little & Rubin, 2019).

No-lookahead and reactivity

Endpoint classifiers, smoothed latent states, reserve proxies, and learned representations may not use future observation availability or terminal labels unless the analysis is explicitly retrospective. The burden and emotional effect of measurement enter the trajectory as intercurrent events. Every cumulative proxy must pass a no-neutrality-by-construction simulation.

Point identification requires positive observation probability or scientifically valid finite bounds across every region of positive phenomenal exposure.

Replicate channels and recoverability

Method-diverse channels can identify or bound a latent distribution only under explicit independence, loading, location, moment, and support assumptions. Two-replicate identification in the spirit of Kotlarski (1967) is available only under its own additive-error and independence conditions; analyst separation and instrument diversity do not prove stochastic independence. Every bridge is first stress-tested in sealed simulations containing assumption violations, shared causes, response shifts, shutdown, and state-dependent reporting. A favorable observed result first constitutes measured-scale evidence and advances to latent PTVC evidence as the protocol-fixed latent gates are satisfied. The Day Reconstruction Method improves structured within-day coverage but remains a retrospective reconstruction whose recall and selection process belongs in the observation model (Stone et al., 2006).

Randomized ecological momentary assessment (EMA) studies show that prompt schedule and response format can alter measured affect distributions or dynamics. Prompt cadence, response format, anchors, and interface defaults therefore belong inside the observation kernel and require frozen or calibrated transport rules (Langford et al., 2026; Haslbeck et al., 2025).

13. Confirmatory decision structure and model recovery

Fix deterministic verdict rules before outcome access. Table 2 is the required confirmatory reporting record: it separates the four-axis public result vector from supporting prerequisite, specification, recovery, and attainment fields, and its conjunctive gates prevent one passed module from masking another’s failure.

Table 2. Required confirmatory reporting record and four-axis public result vector.

Field	Protocol-fixed vocabulary and adjudication rules
PrerequisiteStatus	Valid / failed / unavailable — Representation and sign; neutral origin; protocol-fixed scalar or non-scalar branch; exposure rule and clock; qualifying bearer; ownership and routing; complete strict-pre-endpoint domain; truth-level endpoint; observation and missingness; measured-to-latent bridge. A failed or unavailable prerequisite yields No adequate estimate / invalid gate and blocks downstream promotion.
TargetSpecification / TargetStatus	Type: exact residual / finite band / atom at zero. Modality: [SET-ALL] / [P1] (probability-one) / [POLICY-ALL] / [SUPPORT] / [CONTINUATION-ALL] / [FINITE-CLASS]. TargetStatus uses the controlled verdict lexicon: Equivalent; Different / off-band; Informative but inconclusive; No adequate estimate / invalid gate; or Target-adverse, matched to the specified target. Exact-atom evidence remains nested and branch-matched.
RecoveryStatus	Record generator recovery, scientific-verdict recovery, and absolute adequacy. Recovery or availability failure yields No adequate estimate / invalid gate; failure of the declared open-set check yields Open-set inadequate.
RouteStatus / ComparatorStatus	RouteStatus and ComparatorStatus use the controlled lexicon: Route-favored; Rival-favored / Route-failed; Model-ambiguous; Open-set inadequate; or No adequate estimate / invalid gate. Target-adverse remains a TargetStatus result. Equal-budget scoring, calibration, absolute adequacy, common support, and open-set checks govern promotion.
AttainmentStatus	fixed-policy probability-one / almost-sure attainable / limit-sure only / positive-probability only / unavailable — These route-policy quantifiers do not replace TargetSpecification modality.
InferenceStatus	identified-set / frequentist / Bayesian / sensitivity-based — Preserve the evidence-object type, assumptions, coverage or prior basis, and sensitivity guarantees. Controlled public verdict labels remain on their corresponding axes, with no cross-promotion.

Recovery before interpretation

Synthetic and semi-synthetic studies span PTVC routes, expected-neutrality laws, diffuse and atomic residuals, flexible baselines, conditioned bridges, state-dependent hazards, regulatory and computational models, memory reconstruction, selection, censoring, heavy tails, stochastic volatility, and observation floors.

A correctly computed empty inversion or feasibility set is not NAE and receives its protocol-fixed object-matched state—model-incompatible, bridge-failed, route-failed, or another named rejection. NAE is reserved for unavailable, undefined, invalid, or uncomputed objects. A protocol-fixed recovery matrix reports generator classification and scientific verdict by generator class, including worst-class performance. The pipeline controls false favorable support and false counterexample declarations.

Negative controls and open-set alternatives

Negative-control outcomes, channels, times, endpoint causes, and administrative cutoffs test leakage and common-cause explanations (Lipsitch et al., 2010). Open-set generators prevent a forced choice among misspecified models. If all candidates are inadequate or practically indistinguishable, the correct verdict is No adequate estimate or model-ambiguous—not “best supported.” Proper scores and calibration are reported alongside scientific margins (Gneiting & Raftery, 2007).

Protocol versioning preserves every result: later refinements remain traceable to the version under which each gate and counterexample was evaluated.

Unresolved cases remain visible

Unresolved bearer, endpoint, observation, or bridge cases cannot disappear from the denominator. Use one fixed target, modality, eligible frame, and weighting rule. The displayed interval is a sharp deterministic bookkeeping bound only when favorable and adverse classifications are truth-containing under the same gate system. Confidence-set versions require a simultaneous coverage statement; posterior and sensitivity versions retain their protocol-specified interpretations. Without such a guarantee, report classification bookkeeping rather than a bound on the true violation prevalence.

Transparent unresolved-case bound

$$W_E := W_F + W_A + W_U > 0, \quad \frac{W_A}{W_E} \leq p_{\text{viol}} \leq \frac{W_A + W_U}{W_E}.$$

Let W_F be weighted favorable classified mass, W_A weighted adverse classified mass, W_U weighted unresolved or gate-invalid mass, W_E total eligible mass, and p_{viol} the eligible-population violation fraction. This interval is deterministic bookkeeping, not a confidence or posterior interval unless the protocol-fixed simultaneous inferential guarantee supplies that interpretation.

14. Safe nonterminal program: calibration before terminal claims

The first experiments build the evidential foundation through safe nonterminal studies of representation and local route predictions. Successful modules prepare the program for later complete-stream PTVC testing while initially using nonterminal populations and reversible interventions.

Stage 0 — protocol fixation, public preregistration, and rigorous design

Freeze the target, measurement, route, rival, scoring, error-control, abstention, retirement, and software choices listed in the protocol. Conceal generator labels from analysts and adjudicators until the simulation pipeline is locked. Where roles permit, conceal empirical outcomes until the empirical pipeline is locked. Document the two procedures separately.

Stage 1 — representation and local complementarity

Use safe bounded episodes with randomized benign assignments across valence modalities, controllability, predictability, duration, and horizon, drawing on micro-randomized trial design (Klasnja et al., 2015). Under consistency, well-defined versions, positivity, and the declared interference structure, randomization identifies assignment effects—not delivered, experienced, or latent phenomenal dose. Those dose variables remain separately calibrated with explicit noncompliance and measurement bridges. Test whether an independently measured future response area complements accrued area at randomized cuts against equal-budget ordinary rivals. A transported local cancellation result would support episode-scale route calibration and prepare the next prospectively specified terminal test.

Stage 2 — frozen completion-geometry transport

Estimate one completion potential on training modules and freeze it. On held-out modules, test transition calibration, conditional increments, action odds under randomized information, future-area prediction, residual compression sufficiency, and resolution transport. Off-support actions return partial identification or No adequate estimate. Model selection and representation learning remain inside the training loop.

Promotion criterion

Promotion requires generator and verdict recovery, stable origin and measurement, held-out calibration, superiority to equal-budget rivals, correct negative-control behavior, and replication across a site or channel. Proper scores evaluate predictive distributions; the conjunctive vector remains a decision rubric. Passing supports only the local route and progression to terminal studies.

Immediate analyses in existing data

Before new endpoint studies, existing longitudinal resources can immediately test foundational pipeline components. A measured ledger proxy must be built without future endpoint information. At each prospectively specified landmark, all representations, covariates, competing-risk forecasts, and tuning are fit out of fold using data available by that landmark; realized time and cause are then scored before route-by-surprise tests. Held-out compensator-residual and time-rescaling diagnostics assess calibration under a construction matched to the protocol-specified ordinary, marked, or cause-specific, terminating, and right-censored endpoint process (Cox, 1972; Andersen et al., 1993; Brown et al., 2002). Cause-specific intensity, cumulative incidence or event probability, and interventional endpoint risk are distinct estimands. Thus the test is whether one frozen relation outpredicts selection, local control, prognosis, and hidden-state rivals—not whether hazard merely declines near zero.

Terminal-dispersion analyses must decompose observed variance into latent state, observation error, and their covariance; narrowing can reflect true contraction, censoring, scale compression, sedation, or channel loss. Cohorts with dense affect and later global evaluation can separate experienced value from peak-end, duration-neglect, and narrative-memory rivals. Cross-dataset analyses should retain dataset-specific measurement models and use leave-one-dataset or leave-one-country-out transport rather than force a common coefficient. Causal transport requires explicit invariance assumptions rather than coefficient similarity alone (Pearl & Bareinboim, 2014).

Analyses comparing anticipated with unexpected endpoints freeze a competing-risk forecast at a prospectively specified landmark, score realized time and cause, and only then test route-by-surprise interactions.

Three high-value experiments

First, prospectively specified nonanticipating stopping kernels randomized from frozen history identify or bound occupation functionals only if they separate the protocol-fixed basis. Require well-defined intervention versions, consistency, no interference or a declared interference map, known assignment or conditional exchangeability of future potential outcomes, positivity on common support, and declared observation and censoring models. Report basis, rank, condition number, and null space; distinguish physical stopping from disclosed horizon. Nonseparating or ill-conditioned designs return partial identification or NAE. Second, random-cut complementarity with benign appetitive and aversive micro-interventions separates bookkeeping from causal compensation while assigned, delivered, experienced, and phenomenal dose remain distinct. Third, freeze one completion potential across transition, action, endpoint, and terminal-support modules. All three use sealed simulations, open-set generators, and local claim ceilings.

15. Endpoint-aligned extension and complete-stream adjudication

Observational bridge studies establish the path to terminal tests

Existing longitudinal cohorts and low-burden clinical data can evaluate measurement invariance, time-to-endpoint alignment, missingness models, endpoint revision, residual variance, and rival recovery. These studies create a direct pathway toward PTVC by distinguishing phenomenal closure from symptoms, function, prognosis, and channel shutdown while establishing feasibility, informative bounds, and recoverability.

Prospective endpoint architecture

A terminal protocol needs serial multimethod assessment, an independent endpoint committee, immutable status history, competing-event and return logic, device and communication-quality records, medication and care context, and a joint endpoint–observation–missingness model. The endpoint predicate is fixed without ledger outcomes. Observation burden is minimized and never allowed to interfere with treatment or comfort.

Primary estimands

- The latent terminal residual compatibility set, including origin, endpoint, prefix, suffix, observation kernel, measured-to-latent bridge, and numerical uncertainty.
- Terminal residual atom location and mass versus no-atom-at-zero alternatives, including off-zero atoms, lattice or quantized, singular-continuous, absolutely continuous, and mixed laws.
- Endpoint-dependence and hazard-weighted occupation diagnostics, including unexpected-endpoint stress tests.
- Transport of the frozen completion potential and local signed-dose calibration into the endpoint-aligned module.

Evidence requirements for complete-stream adjudication

Complete-stream adjudication applies the joint prefix–observed–suffix rule from Section 4: the truth-containing object, plus an adjudicated endpoint or validated terminal-remainder bridge, must locate the full latent residual against the protocol-fixed exact or finite-band target. Until then, separate bounds, protocol endpoints, and measured-scale results remain staged inputs.

Clinical ethics govern scientific ambition: studies preserve care and comfort, and no intervention may induce harm, delay care, manipulate dying, or treat PTVC as guidance before empirical adjudication.

Endpoint surprise and promotion

A prospective program distinguishes disclosed horizon, predicted endpoint hazard, and realized time and cause. One route may work only when participants can plan; another may predict structure under concealed or unexpectedly rapid endpoints. Promotion requires the same measurement, recovery, comparator, common-support, and truth-bridge gates used to declare adversity; favorable and adverse claims receive symmetric scrutiny.

16. Exactness, atoms, and symmetric error control

Exact zero is a precise boundary property. Its evaluation separates location, spread, atom mass, and the observation process while accounting for finite resolution, measurement error, smoothing, selection, and model flexibility.

Atom at zero versus no atom at zero statistical branch

$$P_i^Z(dz) = \pi_{0,i}\delta_0(dz) + (1 - \pi_{0,i})G_i(dz), \quad G_i(\{0\}) = 0, \quad 0 \leq \pi_{0,i} \leq 1.$$

For stream i , the no-atom-at-zero alternative G_i satisfies $G_i(\{0\})=0$ and may contain off-zero atoms, lattice or quantized structure, singular-continuous or absolutely continuous mass, and mixtures. The finite-band branch asks whether $P_i^Z([-ε, ε])$ meets a protocol-fixed threshold; the atom branch asks $\pi_{0,i}>0$; probability-one closure asks $\pi_{0,i}=1$. A location-free rival estimates stream-specific atom location a_i and mass $\pi_{a,i}$; its remainder $G_{i,a}$ assigns zero mass to $\{a_i\}$. Atom existence and zero location are therefore scored separately. [SET-ALL] is the stronger deterministic universal claim.

Boundary-aware inference

Both $\pi_{0,i}=0$ and $\pi_{0,i}=1$ are nonregular boundaries, so ordinary interior asymptotics may fail (Self & Liang, 1987); at $\pi_{0,i}=1$ the remainder law G_i is unidentified. Recovery should include parametric bootstrap or valid branch-specific methods, atomless laws with high density near zero, near-unit mixtures, location-free atoms, singular alternatives, rounding, heteroskedastic error, missingness, and finite-sample stress tests.

Nested-resolution discipline

Prospectively nested bands may increase discrimination only while the latent target, acceptance sets, bridge family, recovery guarantees, and limiting rule remain fixed. Passing a smaller observed band after changing the instrument or model does not constitute convergence to exactness. Independent channels and transport across resolution are more informative than repeated analyses of one kernel.

Symmetric error-control budget

The confirmatory multiplicity budget is symmetric: it controls false favorable support across representation, origin, routing, endpoint, missingness, route, rival, tolerance, and subgroup searches, and false counterexample declaration across streams, endpoints, instruments, and models. Adaptive choices remain exploratory or inside training. A protocol-fixed deterministic falsifier, once verified on held-out data with its truth bridge intact, is not weakened by unrelated multiplicity; a noisy search for such a falsifier remains subject to multiplicity control. Promotion and retirement each require their protocol-fixed outer decision set.

17. Falsification, safeguards, ethics, and research agenda

Results that would narrow or reject a protocol-fixed PTVC claim

Because the program is built to learn, clear counterevidence would be a valuable scientific result. A deterministic, nonempty truth-containing residual set for a gate-valid stream that excludes zero refutes the protocol-fixed exact set-universal class. Other branch-matched results include endpoint independence with demonstrably nonzero ledger occupation; imbalance outside validated remaining capacity; reproducible scalar or bipolar representation failure; origin flexibility that manufactures closure; cross-channel incompatibility; and repeated route failure against adequate equal-budget rivals. Frequentist adverse decisions preserve coverage and multiplicity; posterior adversity remains prior- and model-conditional. A better-fitting no-atom model weakens the exact-atom branch; memory-only closure favors memory rivals; missing-not-at-random or endpoint sensitivity can remove terminal support; and adequate-rival parity removes differential route support.

Known challenges and their research solutions

Phenomenal sign, cardinal or interval scale, neutral origin, exposure, stream individuation, ownership, and truth-level endpoint are demanding latent constructs. Complete histories are rarely observed, endpoint and observation processes are entwined, and some models may remain partially identified. The architecture turns these challenges into explicit measurement, partial-identification, transport, and decision problems, allowing each result to retain a clear representation, bearer class, species, culture, endpoint cause, and resolution.

Ethical boundary

Ambitious research is trustworthy only when care comes first. PTVC is presented as a research hypothesis, so current clinical, legal, moral, and policy decisions continue to rest on established evidence and standards. The hypothesis cannot justify exposing anyone to harm, discounting present suffering, denying care, equalizing outcomes, or claiming that another person's experience is compensated. Human research must follow the Declaration of Helsinki, preserve informed consent and withdrawal, protect privacy, minimize burden, and use independent oversight wherever vulnerability or terminal conditions are involved (World Medical Association, 2024). Any animal research must follow applicable welfare and reporting standards and use replacement, reduction, and refinement wherever possible.

What should materially change belief

A strong favorable pattern would combine a frozen valence representation, stable origin and exposure rule, method-diverse channels, one joint truth-containing prefix–observed–suffix object, a validated endpoint and observation model, contraction of the latent compatibility set under linked resolutions, one completion potential transported across modules, superior held-out performance against adequate equal-budget rivals, correct endpoint-surprise behavior, and independent replication. Such a result would constitute strong support for the protocol-fixed class, branch, and resolution while defining the next quantifier-bridge challenge.

These criteria make belief revision constructive: favorable signatures gain force as protocol-fixed bridges pass, while resolved ambiguities narrow the live explanations. Near-zero means, adaptation, terminal decline, cultural recurrence, fitted bridges, and partition slopes gain evidential weight through independent measurement, transport, and equal-budget rival comparison; nonsignificance alone remains nonadjudicative.

Near-term prerequisite and differential predictions

The near-term prerequisite is Section 9's random-cut gate: $\beta = 1$ (slope -1) under independent prefix and remainder calibration. This establishes additivity but not PTVC-specific closure; an independently tested zero intercept, a complete-stream test, and transport across cut distributions supply the discriminating evidence. With $\beta = 1$, a shifted intercept supports nonclosure; a failed slope diagnoses additivity or measurement failure.

Early publishable milestones

A valuable paper can materially advance PTVC by validating one frozen representation, origin rule, exposure clock, observation kernel, missingness model, endpoint interval, or local route in a protocol-fixed class. It should report generator recovery, control performance, uncertainty, and transport. Its immediate contribution is a robust measurement or local mechanism result and a rigorous foundation for terminal studies.

Cumulative decision bookkeeping

Each result enters a versioned evidence ledger recording the target and modality, bearer and routing, representation, origin and exposure, endpoint, observation and missingness, reference law and support, tolerance or atom claim, rivals and budget, estimand and proper score, recovery, decision, and claim ceiling. Outcomes downstream of one perturbation remain one evidence family. Promotion requires its held-out gate; no index offsets a failed target or truth bridge. Replication changes a principal bias—site, instrument family, analyst team, endpoint process, species, or intervention—while transporting the same frozen object.

Promotion, retirement, and open challenge

The project should maintain a public generator bank that includes supportive, rival, mixed, singular, near-zero, measurement-failure, missing-not-at-random, and none-of-the-above worlds. New routes enter only with independent equations and prospective retirement rules. A route retires after reproducible recovery failure, failure of its defining equation under adequate power, or loss to an equally resourced rival across its declared support. A target branch retires when a gate-valid counterexample satisfies its protocol-fixed falsifier. Escalation to vulnerable or endpoint-proximal populations occurs only after safer modules pass, the additional information cannot be obtained otherwise, independent oversight approves the burden, and the protocol cannot influence survival or care. The same infrastructure must make adverse, ambiguous, and no-adequate-estimate outcomes as publishable as favorable ones.

Ordered research agenda

The rational order is: independent formal review and open generator-recovery challenge; nonterminal representation and observation studies; bounded random-cut, signed-dose, horizon, and completion-potential experiments; endpoint-aligned analyses of existing cohorts; prospective low-burden multimethod cohorts with independent endpoint adjudication; and only then direct complete-stream tests. Every stage publishes complete outcomes, versioned protocols, code, formula and claim registries, and predetermined promotion and retirement rules.

Program-level conclusion. PTVC converts a broad conjecture into a typed boundary hypothesis that can be measured, compared, supported, narrowed, or refuted. It brings separable positive and negative affect, opponent and adaptive dynamics, integral feedback, history-sensitive mood, horizon-dependent regulation, terminal trajectories, consciousness assessment, and latent-state and endpoint inference into one testable frame. The program can now determine whether this convergence reflects a terminal boundary law or an explicit rival. A gate-valid counterexample, a transported signature, or a principled No adequate estimate would each advance the science and invite interdisciplinary collaboration, replication, and appropriately staged support.

18. Evaluation and reproducibility standard

Evaluation asks whether the formal target and research architecture meet internal-validity, claim-ceiling, comparative-testability, ethical-feasibility, and reproducibility standards. Meeting them makes PTVC scientifically assessable and creates a disciplined path for empirical support to accumulate through prospectively specified studies.

Modular, noncompensatory evaluation

Evaluation keeps five modules separate: formal coherence; measurement and identification; endpoint and ownership validity; recovery and error control; and ethics and reporting. Strengths and limitations therefore remain local to the affected branch, assumption, estimand, or verdict. A limitation narrows or retires that branch; a blocked complete-stream object receives NAE until its bridge is supplied.

Minimum reproducibility capsule

Every empirical study derived from this framework should archive a timestamped protocol; claim, formula, bearer, routing, endpoint, observation, missingness, generator, data-split, software, seed, and code records. Data provenance, exclusions, transformations, linkage, access restrictions, and time windows must be documented. Outputs should include machine-readable decisions, recovery matrices, acceptance geometry, planned sensitivities, and public-artifact hashes. Privacy or licensing may restrict raw data, but not executable synthetic fixtures, schemas, code, or the information needed to reproduce the decision path.

Independent audit and discrepancy handling

Before outcome access, an independent audit checks object typing, formula order, endpoint and cut randomization, no-lookahead, action domains, policy mass, rank, weighting, multiplicity, and open-set behavior. Corrections are versioned without replacing the original protocol, output, or verdict. Distributed formats and machine-readable sources must match the displayed mathematics and claims. Production fixes leave the scientific model unchanged; scientific changes receive a new version.

Promotion, retirement, and publication outcomes

Publication outcomes follow Sections 14 and 17's protocol-fixed promotion and retirement rules. Every protocol-defined outcome remains publishable; verified counterexamples remain attached to the tested version, finite-resolution compatibility remains resolution-limited, and no target-defining element may move after inspection.

Declarations

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Artwork. Diego Santos.

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Competing interests. Stephen Takowsky originated PTVC, founded the Law of Fairness Project, and owns or controls rights in related publications and project materials.

Data, code, and materials. The synchronized Formal Model (Takowsky, 2026) provides expanded definitions, proofs, protocols, and formula provenance. Empirical studies arising from this framework will release materials under their protocol-fixed ethics, privacy, and version-control plans.

Ethics statement. This work involved no human or animal participants and used no participant data.

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